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Knowledge and the New Zealand curriculum ‘refresh’

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ABSTRACT

This paper examines the extent to which there has been a shift towards disciplinary knowledge in recently developed curriculum documents in New Zealand and evaluates whether a new ‘Understand, Know, Do’ structure for the curriculum has the potential to facilitate coherent design of teaching programmes and ‘deep learning’. Using a social realist lens, Bernsteinian theories on knowledge structures and recontextualization, and the principles of a Curriculum Design Coherence Model, the analysis identifies instances of both conceptual coherence and epistemic confusion which raises questions about the underlying principles upon which the curriculum documents are being developed.

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Introduction

Recently, a number of scholars have noted a new emphasis on disciplinary knowledge in several national curricular developments, for example Bratland and El Gharni (2022) in Norway, Adolfsson (2018) in Sweden, Spielman (2018) in England, and Hoadley (2018) in South Africa. Hughson and Wood (2022) have also noted what they regard as a significant shift in the OECD discourse by the inclusion of some discussion about knowledge in its 2030 Learning Compass (OECD, 2019). This stands ‘in stark contrast to previous OECD reports which have advocated instead for more generic skills and competencies’ (Hughson & Wood, 2022, p. 634). The OECD document states that ‘disciplinary knowledge, or subject specific knowledge, continues to be an essential foundation for understanding, and a structure through which students can develop other types of knowledge. The opportunity to acquire disciplinary knowledge is also fundamental to equity’ (OECD, 2019, n.p.).

In this paper, we ask the question, to what extent is there a perceivable shift towards disciplinary knowledge in the current ‘curriculum refresh’ in New Zealand? The educational context for asking such a question is one where over the last three decades in New Zealand, and in many international jurisdictions, curricula have been characterized by a move away from explicit knowledge identification to an emphasis on outcomes-based curricula, generic competencies, and the cluster of ideas associated with 21st Century Learning such as student-driven pedagogies (Priestley & Sinnema, 2014). Knowledge has been side-lined or taken for granted during this period as it has tended to be equated with an outmoded approach to education, one conceptualized as an induction into a relatively static and narrow repertoire of accepted facts and information. However, in alignment with social realist research about disciplinary knowledge conceived as conceptual and ‘powerful’ (McPhail, 2020c) we argue that the affordances of disciplinary knowledge are of paramount importance for both the cognitive and social development of young people (Young & Muller, 2010) and in agreement with the OECD statement above, that access to this type of knowledge is fundamental to equity. By disciplinary

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knowledge, we mean knowledge that is derived from the disciplines, such as History, Biology, Art etc., that is recontextualized for educational purposes (Bernstein, 2000). Social realism emphasizes the importance of providing access for all students to the conceptual ways of knowing subjects, while recognizing that knowledge is socially constructed and subject to change. We recognize that a pedagogy of active engagement is required to bring knowledge alive for students as it is put to use (Barrett & McPhail, 2021; Rata et al., 2019) but our focus in this paper is on the presence of knowledge in the national curriculum documents presently undergoing revision in New Zealand.

After Popper (1978) we take a realist perspective in our work in that we regard disciplinary knowledge as ‘real’ in the sense that it exists independently from the knower and has effects in the world. It follows from this proposition that differentiating knowledge—the object of learning—from pedagogy—the approach to learning—is helpful in providing a theoretical means to consider the complex interrelated elements of education (Young, 2010). In this paper, we consider the New Zealand curriculum refresh from this social realist perspective drawing on a number of Bernsteinian concepts (Bernstein, 2000) outlined and justified in the ensuing sections of the paper.

Context

In this section, we briefly draw attention to two of the forces that have shaped the direction of international educational discourse at a deeper level over the last three decades: constructivist philosophies and neoliberalism. We suggest that the two discourses can be theorized as now co-located, albeit somewhat uncomfortably, within the broader narrative of 21st Century Learning (Couch, 2018; Robertson, 2016). The influence of these discourses, at least in the New Zealand context, has resulted in ‘a shift in the balance between knowledge and skills/competencies in both the curriculum itself and in discussions of the kinds of educational outcomes thought to be necessary ...’ (Lourie, 2020, p. 124). Lourie argues that this increasing emphasis on skills and competencies at the expense of knowledge can be understood as a response to the knowledge economy discourse which ‘tends to frame education in a predominantly functionalist way’ and enables governments to see more explicitly the results of their investment in education (p. 124).

Most significantly from a knowledge perspective, the discourses of constructivism and neoliberalism have engendered a change in the way in which knowledge itself is understood (Bolstad et al., 2012; Lourie, 2020). For example, some scholars argue that because knowledge dates quickly we do not need to teach it anymore (De Bruychere et al., 2015), that generic skills such as ‘learning to learn’ are seen as more important than what we learn (Claxton, 2013), or that we expect students to construct knowledge themselves (McPhail, 2017; Ministry of Education, 2007, p. 12). Moreover, the curriculum should enable a high level of choice, so learning is believed to be relevant. Such ideas tend to overlook the differences between knowledge as ‘facts and information’ and ‘knowledge as conceptual systems of meaning’, but this differentiation has significant consequences for deep learning (McPhail, 2020b).

The New Zealand curriculum ‘refresh’

The decision to ‘refresh’ The New Zealand Curriculum emerged from a significant change in direction for curriculum design following the publication of the Aotearoa New Zealand’s Histories in the New Zealand Curriculum in March 2022. In 2019, a petition called ‘Give me my History!’, was presented to the government by the New Zealand History Teachers’ Association asking for the teaching of New Zealand history to be compulsory. The New Zealand Prime Minister at that time, Jacinda Ardern, gave her support for this initiative. The idea that there should be a specific History curriculum which identified content that all students should learn went against the approach taken by the Ministry of Education of having a flexible curriculum which left schools and teachers to select content related to broadly stated achievement objectives. Moreover, the new histories curriculum introduced a new structure and new guiding principles which became both the impetus for the development of new

curriculum documents and the prototype for the current curriculum refresh. The refresh began in 2021 and has a planned completion date of 2026. The process for devising the documents includes a high degree of public consultation with Ministry of Education staff leading and guiding expert groups comprising education sector members such as teachers, academics, industry interest groups, and Māori and Pasifika representation.

The Social Sciences curriculum was the first to be ‘refreshed’ (October 2022) and currently there is a draft curriculum—Te Mātaiaho (to observe and examine the strands of learning)—which includes two other learning areas: Mathematics and Statistics and English (Ministry of Education, 2023a, March 2023d). This draft outlines matters such as the ‘refreshed purpose of the curriculum’ (p. 13), a ‘refreshed vision for young people’ (p. 16), and also carries over from the 2007 curriculum elements such as the ‘key competencies’ (pp. 25–6) and ‘values’ (p. 27).

The new Aotearoa New Zealand’s Histories in the New Zealand Curriculum (2022) had signalled a shift to greater recognition that there is content knowledge which is valuable for students to understand and know. Furthermore, the Ministry of Education has stated that the curriculum refresh involves shifting from an outcomes-based curriculum to ‘a progression-focused curriculum’ (Ministry of Education, 2023e, 2023f) through using an ‘Understand, Know, and Do’ framework. The elements of Understand, Know, and Do for each learning area clearly lay out the big ideas, contexts, and practices for the area and enable increasingly rigorous and complex learning’ (Ministry of Education, March 2023d, p. 5).

In contrast to the current ‘concise’ curriculum (see Ministry of Education, 2007) this new model is intended to provide more detail and guidance about what students need to understand, know, and do at various levels of the curriculum. The curriculum documents are structured according to these three components, ‘Understand, Know and Do’, and they appear in both an overview section for each subject and provide an organizational rubric for the more specific ‘progress outcomes’ for each of the new five phases of learning (Years 0–3, Years 4–6, Years 7–8 for Primary and Intermediate schools, and Years 9–10, 11–13 for Secondary schools).

The ‘Understand’ section identifies a number of key ‘Big Ideas’ for each subject: ‘At the heart of each learning area is a unique set of big enduring ideas that all ākonga [students] can relate to and access’ (Ministry of Education, March 2023d, p. 23). The concept of ‘Know’, is fleshed out by reference to more specific disciplinary contexts and/or concepts for learning and teaching, for example number, algebra, measurement, space, statistics, and probability, are identified for Mathematics and Statistics. ‘Do’ outlines the subject practices and skills students might use when applying their knowledge. While there is more explicit acknowledgement of the importance of progression and increased conceptual signalling (Zohar & Hipkins, 2018), the terminology and structure indicate an underlying lack of a clear and logical theory of knowledge that could simplify the way in which the curriculum is structured and make it easier for teachers to follow.

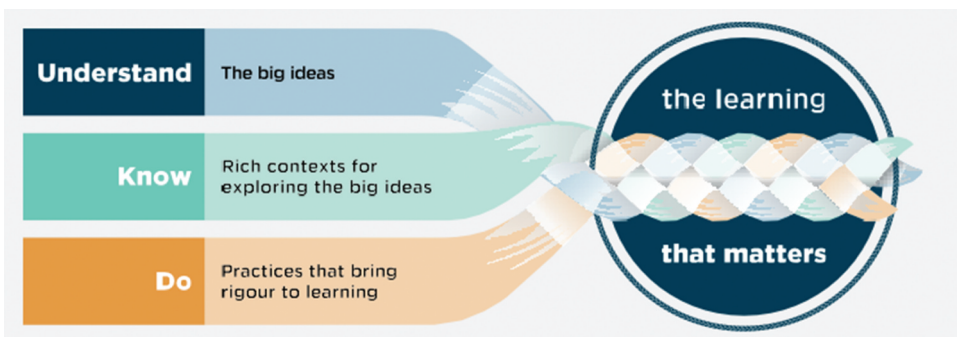


Figure 1. The Understand, Know, and Do Model. From Te Mātaiaho, by Ministry of Education, 2023, <https://curriculumrefresh.education.govt.nz/te-mataiaho>

Theoretical framework

We outline briefly here the socio-epistemic theory of knowledge which we use to undertake our analysis. The theory utilizes work undertaken by realist scholars, in particular Moore (2007, 2013), Muller (2009, 2022), and Young (2008) which in turn draws on ideas from Durkheim (1995), Vygotsky (1986), and Bernstein (2000). From this corpus of work, we identify two key ideas that are operationalized in the Curriculum Design Coherence Model (McPhail, 2020b, 2023; Rata, 2019, 2021; Rata & McPhail, 2020) and which we use in the empirical analysis (see below): (i) the differentiation of concepts, content, and skills and (ii) the interrelationship between conceptual and applied knowledge. These two key ideas are in turn derived from the concepts of (i) knowledge differentiation (the distinction between everyday knowledge and epistemic knowledge)—Vygotsky's 'scientific and spontaneous concepts' (Vygotsky, 1986), and (ii) Bernstein's knowledge structures (Bernstein, 1999). Given that the Curriculum Design Coherence Model is founded on accepted epistemological principles, we consider it an appropriate means to measure the curriculum design of the New Zealand refresh. We draw more on the key principles of the model than on its design particulars. These principles enable us to propose that 'the complex relationship between knowledge types and forms, and between concepts, content, and skills need to be addressed at the curriculum design stage before teaching begins' to enable coherent, conceptual design, and 'deep learning' (McPhail, 2023, p. 42). We define deep learning as the ability to see the connections between epistemic parts and wholes of a subject which in turn leads to the ability to generalize by applying abstract concepts to a range of contexts including the socio-cultural world (McPhail, 2020b; Winch, 2017).

Curriculum design coherence model

The Curriculum Design Coherence Model (McPhail, 2023; Rata, 2021) which Muller (2022) describes as 'a deductive model derived from social realism first principles' operationalizes the differentiation and subsequent necessary connection between conceptual and applied knowledge for curriculum design (p. 11). The Model is based on the premise that the structure of disciplinary knowledge—the parent systems of meaning for school subjects—provides unique affordances for both structuring learning and enhancing human cognitive development through design coherence (Geary, 2012). These affordances are enabled by the bringing together of two forms of knowledge: 'knowledge-that' (propositional knowledge) and 'know-how-to' (applied knowledge) and are a precondition for deep learning (Bratland & El Ghami, 2022; McPhail, 2020b; Winch, 2017). The aim of the Curriculum Design Coherence Model is to enable a deep level of design coherence by aligning curriculum design with the epistemic/conceptual structure of the knowledge to be taught (its hierarchically interrelated abstract and generalizable concepts), content that best materializes the concepts, and learning activities that bring the 'knowledge-that' (concepts and content) together with 'know-how-to' (applied knowledge). Such an approach aims to lessen the possibility of too abstract a curriculum (based on concepts or 'big ideas' only), an overcrowded curriculum (dominated by content such as facts and information), or an overly 'instrumental' curriculum (where skills dominate).

Recontextualization

At a more general level of analysis we draw on Bernstein's (2000) concept of recontextualization, a process whereby specific discourses, such as knowledge produced in universities, are adapted and reshaped as they move from their original sites of production to the curriculum and the school. The recontextualization process 'selectively appropriates, relocates, refocuses and relates other discourses to constitute its own order' (Bernstein, 2000, p. 33). In this process, there is the possibility for original meanings to be misunderstood or altered as ideas and concepts are removed from their original context and juxtaposed with others. There is also the potential for the play of ideology where a particular idea is picked up and applied or emphasized without due diligence to the original body

Table 1. Translation device.

Form of Knowledge	Curriculum Design Coherence Model Components	Indicators	Empirical Examples
Knowledge-that	propositions	statements which propose a main idea	Black Lives Matter is an activist social movement concerned with eradicating systemic racism and upholding human rights.
	Subject concepts	an abstract idea conveyed through a word or short phrase	racism, human rights, social justice, prejudice.
	Content	the material exemplification of concepts which may include facts, contexts, ideology, aspects to know etc.	The Black Lives Matter movement began in the United States and spread internationally. It has brought attention to racial prejudice, injustice, and police brutality as evidenced in the murder of African-American George Floyd on 25 May 2020.
Know-how-to	skills, competencies, disciplinary practices	applying the knowledge through using skills or practices relevant to the discipline	Interpreting primary and secondary sources on specific case studies of purported systemic racism and applying human rights principles in a range of contexts.

of knowledge. We look therefore for evidence, either explicit or implicit, of the *recontextualising principles* drawn on by the Ministry of Education in the curriculum refresh documents. For example, Wheelahan (2010, p. 124) theorizes that instrumentalism and relevance, rather than ‘systematic access to structures of knowledge’ has dominated educational developments in the last decades within what Bernstein terms the Official Recontextualizing Field, the field dominated by government or international agencies such as the OECD (Lourie, 2020). We theorize the recontextualizing principles drawn on in the curriculum refresh in Part 1 of our findings below.

Regulative and instructional discourses

As part of the recontextualization process, Bernstein (2000) identifies two mechanisms or principles used to construct a curriculum: an instructional discourse and a regulative discourse. An *instructional discourse* comprises the selected knowledge, its organization through selection, sequence, pacing, and evaluation. The instructional discourse is embedded in a stronger *regulative discourse*, ‘the moral discourse which creates order, relations and identity’ (Bernstein, 2000, p. 32). The regulative discourse functions ideologically and the dominant principles of those in control of the ‘official recontextualizing field’ are contained within the regulative discourse of the curriculum (Bernstein, 2000). We utilize these concepts in our analysis below.

Methods and data analysis

Utilizing the principles of the Curriculum Design Coherence Model, as outlined above, we developed a translation device (Pountney & McPhail, 2017) or analytic rubric (see Table 1) based on that developed by Bratland and El Gharni (2022) for their study of Norwegian curriculum reform. Using the rubric, in Part 2 of our findings, we examine two key aspects of the refreshed New Zealand Curriculum (Ministry of Education, March 2023d): the identified purpose for the refresh (pp. 12–13) and the overall structure—in particular the concept of learning progressions (pp. 17–19) and the ‘Understand, Know, Do’ structure (p. 23). We then look in more detail at the ‘Progress Outcomes’, using the outcomes for the end of Year 10 (the second year of secondary school), as an example, for Social Sciences, English, Mathematics and Statistics, and for Aotearoa New Zealand’s Histories.

Findings

Part 1 Curriculum principles in the ‘refresh’

Ahead of the publication of the current 2007 *New Zealand Curriculum* document, Clark (2004) raised concerns about the Ministry of Education’s approach to curriculum development. He describes the Ministry of Education’s approach at this time as one lacking ‘a basic over-arching central philosophy’, rather it was described by Kelly (cited in Clark, 2004, p. 130) as one of ‘rigorous eclecticism’. Clark argues that the eclecticism was visible in the way the curriculum drew on approaches with quite disparate epistemologies such as postpositivist realism, instrumental rationalism, postmodernism, progressivism, and constructivism which can create underlying contradictions and tensions. In summary, he and other commentators argued at that time that curriculum development in New Zealand suffered from ideological confusion and the overriding influence of economic rationalism or instrumentalism (Boyask, 2004; Mansfield, 2003; Peters, 2003).

Twenty years on we can see the Ministry of Education utilizing mātauranga Māori (Māori knowledge) as a means to create a sense of general thematic and metaphorical cohesion throughout the curriculum refresh, through the use of Māori terms and philosophical ideas. For example, the following quotes indicate this primary focus: ‘Te Mātaiaho is designed to be a curriculum that gives effect to Te Tiriti o Waitangi’¹ (Ministry of Education, March 2023d, p. 5); ‘The curriculum takes its direction from Te Tiriti o Waitangi and its principles, particularly its provision for the active protection of taonga,’² including te reo Māori,³ tikanga Māori,⁴ and mātauranga Māori⁵ ...’ (March 2023d, p. 8); the curriculum ‘aims to transform educational inequities for Māori, giving prominence to mātauranga Māori and te reo Māori’ (March 2023d, p. 12).

This utilization of mātauranga Māori helps create a strong bi-cultural *regulative discourse*: ‘the moral discourse which creates order, relations and identity’ (Bernstein, 2000, p. 32). The regulative discourse functions symbolically in that it tells us what should be valued, as well as projecting an imaginary of the future. Bernstein (2000) also theorizes that an *instructional discourse* (what is taught and how it is taught) is embedded within the stronger regulative discourse. The *recontextualising principles* reproduce and recontextualise the dominant discourses and ideologies present within a society at a particular time. For example, Siteine (2018) has theorized that a strong ‘identity directive’ has been encoded in New Zealand curriculum discourses for some time—a directive derived from a purported ‘causal link between achievement and recognizing ethnic identity using the cultural values, language, and practices in the classroom to achieve this’ (p. 57). This identity directive is strengthened in the refresh.

The shift towards recognition of Te Ao Māori (the Māori world view), mātauranga Māori, and the Te Tiriti o Waitangi has its roots in both political and educational discourses, in particular the research that purports that success for indigenous learners requires cultural recognition within both the regulative and instructional discourses of the school and culturally responsive pedagogies in the classroom (Berryman et al., 2018; Bishop, 2019; Hargreaves, 2022). It is clear in the following statement how the curriculum refresh also draws on particular curriculum theories to highlight regulative goals rather than instructional learning goals such as those that might be enlisted from epistemology or cognitive science (e.g. Sweller, 2021; Winch, 2017). In other words, the agents of recontextualization appear to be seeking evidence to support their ideological position that foregrounds culture and heritage:

The curriculum is informed by the knowledge landscape and by insights from *a range of curriculum theories* that are transforming our understanding about the importance of our dual heritage, our culturally diverse communities, and a shared, sustainable, and prosperous future for Aotearoa New Zealand. (Ministry of Education, March 2023d, p. 12, italics added)

The educational literature that has been drawn on to develop these ideas is not clear. While some authors are cited in associated Ministry documents (e.g. Ministry of Education, 2022), particular

references are not provided in Te Mātaiaho to explain the overriding discourses of the curriculum, in particular how learning happens.

It is not our intention to provide a complete analysis of the variety of recontextualising principles ‘bumping up’ against each other in Te Mātaiaho, but our analysis points to a continued eclectic approach that attempts to integrate quite disparate epistemologies, theories, and ideas despite the overarching use of concepts from mātauranga Māori. For example, in relation to the recontextualising principle of biculturalism, we note a related ‘localizing ideology’. The principle of localization foregrounds ‘designing and reviewing school curriculum with local communities’ (Ministry of Education, March 2023d, p. 5) and ‘learning through local relationships with tangata whenua (people of the land) and the community’ (Ministry of Education, March 2023d, p. 6). While making the boundaries between the school and its communities more permeable is likely to engender many positive outcomes as a *regulative* ideal or principle it is possible it may create a tension point if the concept overrides the *instructional* ideal through influencing significant portions of curriculum content. For example, what eventuates if local ideas and priorities come into tension with the principle of ‘learning that cannot be left to chance?’ (Ministry of Education, March 2023d, p. 5). Such learning is stated as key ideas and understandings specified at different levels of schooling in the ‘progress outcomes’ (Ministry of Education, March 2023d, p. 18) but the recontextualization and localization of curriculum in schools could challenge these progress markers. The principle of high expectations as described here may also contradict the strong presence of localization:

Leading kaiako [teachers] to take ākonga [learners] beyond their immediate experience, inspire new curiosities, and open up new horizons, so that ākonga develop both broad and specialized knowledge and grow and excel as local and global citizens with a contribution to make (Ministry of Education, March 2023d, p. 15).

Moreover, our analysis points to a potential contradiction between the stated primary focus of the curriculum of incorporating mātauranga Māori in realizing the principles of Te Tiriti o Waitangi and the claim that the curriculum has been *repositioned* ‘to focus on progression and growth in learning, as well as provide more clarity (than the previous curriculum levels) about what is expected for each phase, in terms of both learning and the teaching focus’ (Ministry of Education, March 2023d, p. 18). However, localization is also an outcome of a curriculum aimed at avoiding over prescription where decisions concerning specific content are made at the school level. It remains to be seen how teachers will manage to bring together the aspirational, regulative aims of mātauranga Māori and the knowledge progressions which are based on quite different ontological and epistemological underpinnings (Hikuroa, 2017; Stewart, 2022).

We also note the strong presence of moral principles in the curriculum refresh (Regelski, 2021). ‘Deontological’ ethics guide and assess our choices of what a curriculum should provide, for example: ‘Te Mātaiaho ... is a curriculum designed for all ākonga and their right to belong and flourish through high-quality learning experiences’ (Ministry of Education, March 2023d, p. 5) and virtue ethics provides indicators of what kind of students we want to create: ‘ākonga [learners] develop both broad and specialized knowledge and grow and excel as local and global citizens with a contribution to make’ (Ministry of Education, March 2023d, p. 15). The recontextualising principles of biculturalism, localization, and ethics are three of the most prominent in our analysis and each creates potential ontological and epistemology tensions which are either not understood or avoided through a bricolage approach in the design of Te Mātaiaho.

Part 2 Analysis of knowledge in the curriculum refresh

In exploring the extent to which there is a shift towards disciplinary knowledge in the curriculum refresh in New Zealand, the findings in this section are discussed and organized in terms of the elements which contribute to creating a coherent curriculum design as theorized in the Curriculum Design Coherence Model which in turn are derived from the epistemological literature. These elements are ‘knowledge-that’ (propositions, subject specific concepts, stated content and/or

learning contexts), and ‘know-how-to’ – the application of knowledge (see Table 1, translation device). The analysis identifies the distinctions between these forms of knowledge within the ‘Understand, Know, Do’ structure of the curriculum refresh and considers whether the curriculum emphasis on weaving together the three components ‘Understand, Know and Do’ achieves the stated curriculum goal of ensuring access to ‘learning that cannot be left to chance’ (Ministry of Education, March 2023d, p. 5). Our analysis also considers the extent to which teachers and learners have certainty about what will be learnt, how ‘easy’ (Ministry of Education, March 2023d, p. 5) the curriculum is for teachers to implement, and the extent to which teachers have autonomy over what is taught (Ormond, 2017).

Knowledge-that

The ‘Understand—Big Ideas’ sections refer to the overarching ideas for each particular learning area (a learning area such as ‘The Arts’ may comprise more than one subject) and fall loosely within the ‘knowledge-that’ category in the Curriculum Design Coherence Model. However, these ‘big ideas’ are differently conceived and constructed depending on the subject. In the *Aotearoa New Zealand’s Histories in the New Zealand Curriculum* the ‘big ideas’ in the ‘Overview’ section, as intended, broadly capture knowledge that should be learnt, for example:

The course of Aotearoa New Zealand’s histories has been shaped by the use of power. Individuals, groups, and organisations have exerted and contested power in ways that improve the lives of people and communities, and in ways that lead to exclusion, injustice and conflict. (Ministry of Education, 2022, p. 2)

This ‘big idea’ identifies the key subject concepts of ‘power’, ‘injustice’, and ‘conflict’ which can then be understood by students through applying the concepts to specific examples, such as the dominance of the British Crown during the post Te Tiriti o Waitangi (1840) colonization period which unjustly excluded Māori from their traditional lands.

On the other hand, the ‘big ideas’ for Mathematics and Statistics tend to be expressed in terms of socio-cultural conceptions of mathematical knowledge, which often serves as a justification for the value of learning Mathematics and Statistics rather than indicating what should be learnt.

Mathematics and statistics have a continuous, evolving history

Mathematics and statistics have been constructed over thousands of years across the globe as we have grappled with notions of quantity, numerical representation, measurement, dimension and pattern. They continue to be constructed from ideas from many cultures. In Aotearoa New Zealand, our location in Te Moana-nui-a-Kiwa [the Pacific Ocean]— with its multiple cultures, artefacts and knowledges—contributes to mathematics and statistics (Ministry of Education, May 2023b, p. 5).

In this example, the nature of the knowledge is expressed in terms of its ongoing evolution and the value of different cultural understandings of Mathematics and Statistics is advocated. While this ‘Big Idea’ references core concepts such as ‘measurement’ and ‘quantity’, they appear to be included only as an indicator that they are inherent in the knowledge which has been produced and evolved, rather than being included for the purposes of indicating concepts and related content that should be taught. This ‘Big Idea’ suggests that the developmental history of Mathematics and Statistics will be taught yet the contexts stated in the ‘Know’ section, while referring to different cultural approaches, does not include content about the historical development of Mathematics. As such, there seems to be a disconnect between the ‘Big Idea’ and the content used to elaborate it; this ‘Big Idea’ provides few clues for teachers about what ‘knowledge-that’ should be taught and ‘not left to chance’. The teacher has to do the work of connecting the big ideas to the subject concepts to be taught in the know section.

The ‘Big Ideas’ for English comprise statements which both identify ‘knowledge-that’, such as ‘Communication depends on shared codes and conventions’ (Ministry of Education, May 2023a, p. 4) and those which argue the case for the importance of the subject, for example,

Stories are a source of joy and nourishment. Enjoying the stories of others and crafting our own provide us with opportunities to experience different worlds through creativity, imagination, and interaction. ... Broadening and deepening the intellectual and aesthetic appreciation of story is a worthwhile outcome in itself. (Ministry of Education, May 2023a, p. 4)

The 'knowledge-that' in this 'Big Idea' is that students will be deepening their intellectual knowledge about stories. However, much of the idea is more of a justification for the subject's inclusion in the curriculum, for example the aspirational suggestion that students will develop their imaginations and creativity along with the aforementioned 'joy and nourishment' anticipated in studying and creating stories. These arguments for why this aspect of English should be included, would seem to be better suited to the section which precedes the 'Understand—Big Ideas' section entitled 'Purpose statement for English' (Ministry of Education, May 2023a, p. 2).

Propositions

Propositions are statements which encapsulate a key idea or 'fact'. They therefore serve to provide a clear identification of the aspect of *knowledge-that* which is to be taught and learnt by claiming that 'something is the case' (Ryle, 1946). While the term 'proposition' is not used in the curriculum refresh, propositions nevertheless appear frequently in the curriculum documents and serve a useful purpose. They appear in both the 'Understand' sections, as part of the grammatical structuring of many big ideas, and in the 'Know' sections. For example, the proposition 'Literature, language, and texts embody power relationships' (Ministry of Education, May 2023a, p. 4) is included as a 'Big Idea' while another proposition, 'Texts are shaped for particular purposes with particular audiences in mind' (Ministry of Education, May 2023a, p. 5), is included in the list of things to 'Know'. An example of a proposition from Social Sciences—'Movement within and across borders impacts on people and places. Interactions change people's culture and identity, communities and countries'—is a matter to 'Know' (Ministry of Education, May 2023c, p. 20).

However, in the 'Know' sections for the different year levels for Mathematics and Statistics there are, unlike other subjects, two columns entitled 'I know' and 'I know how to'. Propositions suitably appear in the 'I know' column, such as for Years 9 and 10, 'Multiplying a fraction by an equivalent form of 1 results in an equivalent fraction' (Ministry of Education, May 2023b, p. 24) whereas the 'I know how to' statements are not stated as propositions, for example 'simplify fractions using the highest common factor' (Ministry of Education, May 2023b, p. 24). This inconsistency in the framing of *knowledge-that* as propositions or other forms of statement sometimes illustrates a blurring of the boundaries between different forms of subject knowledge, that is between propositional knowledge ('knowledge-that') and applied knowledge ('know-how-to'). So in this instance there appears to be an unusual inclusion of 'know-how-to', or applied knowledge, in a section intended to identify suitable contexts which 'enable the illumination of these big ideas' (Ministry of Education, March 2023d, p. 23). In the corresponding 'Do' section for Years 9 and 10 the statements are more generalized such as 'I can pose a question for investigation' or 'I can use representations to learn new ideas, explain ideas to others, investigate conjectures, and support arguments' (Ministry of Education, May 2023c, 2023b, p. 27). This crossover between the categories 'Know' and 'Do' points to epistemological confusion. It appears therefore that in this development phase of the curriculum, inconsistencies have arisen and some of these have been observed and addressed while others, such as the conflation of 'knowledge-that' and justifications for the subject being included in the curriculum have not, as yet, been noticed.

Subject concepts

Subject concepts, a key element in the Curriculum Design Coherence Model, enable students to understand, articulate, and make generalizations about a phenomenon. Concepts enable more abstract thinking and should lie at the heart of curriculum design if deep rather than surface content

learning is the intended outcome (McPhail, 2020b). Analysis of the curriculum refresh documents suggests that while concepts can be found throughout the 'Understand' and 'Know' sections they are rarely made explicit, so their significance is likely to be overlooked. While not identified clearly, specialist readers can make suppositions about what the concepts are, and which concepts are intended but these may be interpreted in different ways. This means that the intent of the knowledge statements is not clear enough and teachers may interpret the key ideas for learning in very different ways. This might go against the aim that learners will engage in 'learning that cannot be left to chance' (Ministry of Education, March 2023d, p. 5). In Social Sciences for Years 9 and 10, for example, in the 'Understand' section, 'colonization' is a key concept in the proposition 'Colonization and settlement have been central to Aotearoa New Zealand's histories for the past 200 years', but readers may or may not put equal emphasis on 'settlement' (Ministry of Education, May 2023c, p. 20) as a key concept. This could be significant as a lack of signposting of how colonization led to settlement and are interrelated, in other words the inferential relationship between the two concepts may shift the balance between teaching about colonization and teaching about settlement i.e. the emphasis on that is taught. Similarly, in the 'Know' section the proposition 'Māori as tangata whenua [the people of the land] were excluded from these cultural ideals, which they experienced as colonizing and assimilating' (Ministry of Education, May 2023c, p. 20) refers again to the concept 'colonization' but also includes the concepts 'tangata whenua' and 'assimilation' and the emphasis on these concepts may differ significantly from one school to the next leading to differences in what students may learn. Nevertheless, both these examples, which are propositions, illustrate congruence with the Curriculum Design Coherence Model in that teachers may select content to illustrate the concepts contained in the propositions in ways which enable both specific knowledge of, for example the colonization of New Zealand, and the concept of colonization as a more abstract concept.

In some cases, however, the conceptual underpinning for the curriculum propositions is less evident. For example, the proposition 'Aotearoa New Zealand's relationships with Pacific states since the Second World War have reflected its own interests. These have coincided at times with the interests of Pacific states' (Ministry of Education, May 2023c, p. 21) specifies the Pacific and the period after World War II as the context but does not contain key subject concepts. The proposition could have included concepts such as 'independence' or 'bilateral relations'. Nor does this statement provide guidance on the content to be taught, such as which Pacific states and what 'interests' are intended. It is therefore unclear how 'the proposition connects the topic to the subject concepts, thereby beginning the process of design coherence' (Rata, 2021, p. 466).

Content

Element 2 of the Curriculum Design Coherence Model identifies the important differentiation and connection between subject concepts and subject content. Content (sometimes referred to as 'contexts' in the refresh documents) is the material exemplification of concepts which may include facts, ideas, and contexts. In the curriculum refresh, subject content largely falls within the 'Know' section which is described as 'rich contexts for exploring the big ideas' (see Table 1). An analysis of the 'Know' sections for each of the subjects which have been developed so far suggests that individual subject writing panels are making decisions about the specificity of their stated 'contexts' for learning. For example, the curriculum draft for Mathematics and Statistics is more specific and detailed than either English or Social Sciences. The reduction in specificity appears to suggest that the Ministry of Education is pursuing a broader approach in the curriculum refresh statements and that extensive autonomy over selections of content is being given to teachers, rather than favouring specificity. If this is the case, then it could be difficult to ensure that what is learnt is consistent across New Zealand.

In the Social Sciences curriculum (Ministry of Education, May 2023c) the 'know' sections are clearly subject knowledge and are framed as ideas which then need to be unpacked to determine particular

content and contexts to explain the 'know' propositions in sufficient depth for students to make sense of them. For example, 'Climate change and environmental degradation are impacting inequitably on different communities' (p. 21). However, there is inconsistency in that some of the statements offer examples while others do not. The following proposition requires teachers to select particular times when the Crown asserted its power, or particular occasions when Māori may have taken protest action to 'renegotiate the colonial relationship'.

The Crown asserted its power to establish a colonial state that in consequence diminished mana Māori. Over time, Māori have worked inside, outside, and alongside the Crown to renegotiate the colonial relationship with the Crown and affirm tino rangatiratanga [self-determination, sovereignty, self-government] (Ministry of Education, May 2023c, p. 21)

The time frame, the level of detail or the breadth, and extent of case studies are not suggested.

In other statements examples are given which add a layer of certainty for the teacher about the intent and requirements of the knowledge to be taught:

At different times, various groups have been marginalised in Aotearoa New Zealand. These groups have sought to remedy injustices and practices (e.g. through the Disability Action Group, the Polynesian Panthers, and petitions to governments). Governments have sometimes acknowledged these injustices (e.g. through the poll tax apology and the apology for the dawn raids). (Ministry of Education, May 2023a 2023c, p. 20)

In the finalized document for the *Aotearoa New Zealand Histories Curriculum* 'key questions' and 'learning experiences' further flesh out the 'knowledge-that' and while these are not intended to be prescriptive, they do provide teachers with considerable guidance on what could be taught. In the statement above there are details about specific groups who were marginalized and discriminated against. The curriculum documents for Mathematics and Statistics and English, and Social Sciences, currently in their draft form, do not have 'key questions' and 'learning experiences' sections.

Know-how to

The Curriculum Design Coherence Model advocates for the connection between 'knowledge-that' and 'know-how-to' to be explicit. The justification for this is that, as Ryle (1946) notes, 'effective possession of a piece of 'knowledge-that' involves knowing how to use that knowledge, when required, for the solution of other theoretical or practical problems' (p. 16). What in the Curriculum Design Coherence Model is termed 'intelligent know-how-to' and 'knowing-why' occurs when a student can demonstrate and articulate the connection between the two knowledge forms. These connections are an indicator of deep learning rather than surface learning such as recitation of facts or formulaic demonstration of skills. The connection between the various forms of knowledge is *implied* in the curriculum refresh where the components are intended to be woven together when taught (see Figure 1). What one is required to 'Know' and what one is required to 'Do' is however shown as separate components. For English, the proposition that *'Text creators choose text structures to create meaning. These structures can be combined and recombined for particular effects'* (Ministry of Education, May 2023a, p. 14) could be illustrated to students when they act according to the 'Do' statement to *'deliberately select from a range of strategies ... to make meaning from texts ...'* but they are not connected (Ministry of Education, May 2023a, p. 15). While it is important that teachers focus on the links between 'knowledge-that' and 'know-how-to' as it is desirable in creating a coherent curriculum (McPhail, 2020b; Winch, 2017), the teacher has to work quite hard to make the potential lesson design connections between the statements in each section.

Table 2. Comparison of the New Zealand curriculum 2007 and Te Mātaiaho for social sciences.

The New Zealand Curriculum 2007 Achievement Objectives	Te Mātaiaho May 2023 Social Sciences 'Know'
Students will gain knowledge, skills, and experience to: • Understand how people's management of resources impacts on environmental and social sustainability. • Understand how people seek and have sought economic growth through business, enterprise and innovation.	• The uneven acquisition and allocation of scarce resources, goods, and services, and wealth creates personal, societal, and global challenges. These challenges lead to individual and collective action. • Through innovation and enterprise, individuals, communities, and societies create new opportunities. These opportunities can enrich or damage lives, challenge lives about needs and wants, and impact on sustainability.

From *The New Zealand Curriculum*, 2007 and *Te Mātaiaho*, May 2023 (p. 21) by the Ministry of Education.

Discussion

A move to knowledge

At this point in time the curriculum refresh is partially complete with many subjects yet to be developed. Nevertheless, there is sufficient evidence to draw conclusions on the extent to which the curriculum documents, such as the finalized Aotearoa New Zealand Histories Curriculum and the draft Te Mātaiaho, demonstrate a shift towards greater emphasis on disciplinary knowledge in ways which make knowledge structures visible and ways in which coherent, rather than fragmented learning is the likely outcome. The ideological underpinnings for the curriculum refresh are also evident at this point, although future analyses have the potential to reflect further on matters such as whether the curriculum refresh can reasonably be viewed as grounded in well-recognized theories of knowledge (Muller, 2009; Ryle, 1946; Winch, 2013), whether the effects of the forces of market neo-liberalism and instrumentalism have been significantly mitigated, and whether localization or nationalization of curriculum decisions dominates in the enacted curriculum.

Following a social realist line of argument, we are looking for whether the knowledge identified in the 'Understand, Know, Do' structure sufficiently recognizes, includes, and provides some guarantees that the concepts and content which are likely to lead to students having access to 'powerful knowledge' (Muller & Young, 2019) are included in the curriculum. While some of the 'Big Ideas' encapsulate critical knowledge, as evidenced in the aforementioned example from the Aotearoa New Zealand Histories curriculum discussing how histories have been 'shaped by power', sometimes the 'Big Ideas' emphasize why the subject is important rather than what knowledge is important. Similarly, the inclusion of concepts is somewhat haphazard with some of the 'Understand' and 'Know' statements including concepts, which enable generalization from the particular to more abstract powerful thinking, while other statements are devoid of concepts.

To evaluate whether this is a significant shift towards knowledge in the mandated curriculum for New Zealand, these developments need to be measured in relation to the current *New Zealand Curriculum* 2007. For Social Sciences for example, for Years 9 and 10 (age 13–15), in both the 2007 curriculum and the 'curriculum refresh', there are statements related to 'Economic activity' (see Table 2). The 'refreshed curriculum indicates a shift to *propositions*, rather than the *outcomes* of the 2007 document. The propositions provide an interpretation of the likely core content but the content is not extensive and may provide few guarantees that suitable or adequate content will be addressed. There is a slight increase in the length of the statements being used for similar ideas.

For English, using Year 10 as a comparative example, we see greater development of the aim to look at the purposes and audiences for text (see Table 3). Rather than additional indication of content, there is a greater explanation of how texts can be interpreted in different ways and how texts need to be read critically for their bias. This should provide greater understanding among teachers on what is required but it does not go so far as to specify the types of text or name them. In this case, it could be considered to be 'learning that matters' but perhaps not 'learning that cannot be left to chance'.

Table 3. A comparison of the New Zealand curriculum 2007 and Te mātaiaho 2023 for English.

The New Zealand Curriculum 2007 Achievement Objectives 'Purposes and audiences'	Te Mātaiaho May 2023 English 'Know' 'Text purposes and audiences'
Students will: Show an understanding of how texts are shaped for different purposes and audiences. • Show an understanding of how texts are shaped for different purposes and audiences.	I know: All text creators have biases, which may undermine or enhance their intended purposes. These biases may not be immediately apparent to them or their audience. Audiences differ by time and place. This means they are influenced in different ways and may interpret a text differently from one another and from me. All text creators have biases, which may undermine or enhance their intended purposes. These biases may not be immediately apparent to them or their audience. Audiences differ by time and place. This means they are influenced in different ways and may interpret a text differently from one another and from me.

From *The New Zealand Curriculum*, 2007 (Level 5 Achievement Objectives for English) and *Te Mātaiaho*, May (2023) (p. 14) by the Ministry of Education.

Localization

In these two examples from Social Sciences and English, recontextualization of the knowledge from its place of origin, commonly within universities, to schools, is therefore recontextualization by the teacher or department within a school rather than substantial recontextualization occurring at the locus of the national curriculum. The *Aotearoa New Zealand Histories Curriculum* on the other hand has a higher incidence of specific content being stated in ways which suggest that there would be greater commonality in teaching this knowledge, e.g. 'The Waitangi Tribunal investigation process and subsequent settlements by the Crown ...' (Ministry of Education, 2022, p. 25). The greater specificity of content may also be knowledge which is likely to be closer in proximity to its original site of production such as knowledge regarded as reliable within Māori and/or academic communities. This would suggest that current changes to the curriculum could be considered to be an important shift towards a social realist notion of verifiable powerful knowledge (Ormond, 2019). Nevertheless, a focus on localization of the curriculum along with the broad framing of many curriculum statements is prevalent throughout the curriculum and has the potential to undermine the stated aim of identifying 'learning that matters' and 'learning which cannot be left to chance'. The curriculum 'is designed with the expectation that schools will draw on local content to give effect to the national curriculum' (Ministry of Education, March 2023d, p. 38). Schools are required to 'engage in an ongoing process using the national curriculum to design a rich and meaningful school curriculum ... In local partnerships with tangata whenua (the people of the land), ākonga (students) whānau (families) and the wider community' (p. 38).

The expectation that the design of school-based curriculum should account for the preferences of various communities may pose challenges in prioritizing one community over others and result in complexities when determining the content to be taught. Furthermore, what students learn at school could widely differ from one locality to another which provides few guarantees that the learning will be equitable and robust. Being assured that student learning will include the most critical knowledge, which enables them to engage in the world beyond their communities, will be challenging.

Recontextualizing principles and the regulative discourse

The constructs of the regulative discourse embedded in the curriculum refresh documents, rather than being the neutral carrier of certain expectations, are themselves implicated as part of the unavoidable ideological mechanism inherent in the construction of any curriculum (Bernstein, 2000). We argue that there is no easy escape from the play of ideology. Bernstein suggests that whoever or

whatever dominates in the construction of the field acts as 'a symbolic regulator of consciousness' (p. 37). One aim of our analysis is to make the nature of the components of the regulative discourse more explicit, so the ideas are available for discussion and contestation. This surfacing of key ideas is a means through which one can determine if recontextualizing principles become ideological in the negative sense of the word, as denoting a form of false consciousness for those involved in education to follow. For example, much of the critical discussion surrounding the development of the 2007 New Zealand curriculum, and subsequent policy changes, noted the undeniable influence of neo-liberalism through the concept of a 'knowledge economy' (Harris & Ormond, 2019; Lourie, 2018, 2020). This recontextualizing principle is evident in mechanisms such as outcomes-based curricula, a prioritizing of generic skills and competencies, and in the emphasis on relevance and student choice that are now mostly taken as normative dimensions of education (Lee et al., 2004; Lourie, 2020; McPhail & McNeill, 2021). While most of these mechanisms remain in the education structures in New Zealand, there is less explicit reference to neoliberal concepts such as 'enterprising and entrepreneurial' and being 'contributors' to the economic 'well-being of New Zealand' (Ministry of Education, 2007, p. 8). There appears to have been a shift away from explicit mention of market values to a more global, sustainability emphasis.

We have also noted how biculturalism and deontological and virtue ethics dominate the regulative discourse of the curriculum refresh, to develop a certain imaginary of a type of citizen. The primary question of exactly what we, as a nation, expect our students to know, do, and understand at school still appears to be a secondary or subsidiary concern in the curriculum. Acknowledgement of the importance of subject conceptual knowledge and epistemic signalling and ascent (Winch, 2013) is missing from the regulative discourse of the curriculum. From a social realist perspective, we argue that knowledge itself is the pivotal element in the role of a school and that which differentiates school education from other social institutions. What knowledge students have the right of access to is what a curriculum should shed light on. We suggest that this lack of clarity about *what* should be taught at school is part of the obstinate problem of underachievement in education systems (Muller & Hoadley, 2019).

While the curriculum refresh provides more explicit acknowledgement of the importance of conceptual progression (Rata, 2016) or epistemic ascent (Vernon, 2020; Winch, 2013) and increased conceptual signalling (Zohar & Hipkins, 2018) through the development of new progress outcomes, the instructional discourse terminology and structure indicate an absence of a logical underpinning theory of knowledge that could simplify the way in which the curriculum is structured and make it easier for teachers to follow. For example, we argue that the new structure of 'Understand, Know, Do' (Ministry of Education, March 2023d, p. 23) appears to be a reference to the seminal epistemological literature (e.g. Vygotsky, 1986) but it remains somewhat confused from both epistemological and psychological perspectives. Despite the claim that 'these elements are not separate, and they are not in sequence' and that they should be woven together (Ministry of Education, March 2023d, p. 23) we argue that in both a curriculum design context and a learning context 'Know, Do, Understand' is a more logical conceptual frame than 'Understand, Know, Do'. However, by 'understand' we mean the demonstration of student learning through both applied knowledge and the ability to explain and justify understandings in relation to what has been learned earlier. This is quite different from the conception of understanding used in the curriculum refresh, which, we argue, is somewhat loose and often a distraction in relation to 'learning that cannot be left to chance'. The 'Big Ideas' are often broad meta-statements not sufficiently connected to a subject's epistemic structure to provide the necessary conceptual signalling for coherent design. In the design phase, teachers first need to be clear on *what* is to be known ('Know') before that knowledge is put to work in some way with students ('Do') and then learning is demonstrated when 'knowledge-that' ('Know') and 'know-how-to' ('Do') come together to demonstrate what has been understood ('Understand') (McPhail, 2023; Ryle, 1946; Winch, 2017). It is our assertion that there is a lack of logic and cohesion in the curriculum documents because there is no clear underlying epistemological 'system of meaning' guiding the design of the curriculum.

Conclusion

Overall, our analysis suggests Te Mātaiaho is a document concerned primarily with creating a regulative discourse—a discourse of ethical and political obligations more so than what might be expected in a ‘traditional curriculum’. In this regard, there is not a shift towards disciplinary knowledge. However, the inclusion of new knowledge progressions and the structure of ‘Understand, Know, and Do’ represent attempts to increase knowledge signalling in the new curriculum and is a shift away from outcomes. While it is claimed that the structure is ‘clear, and easy to use’ (Ministry of Education, March 2023d, p. 5) we as academics found it to be complex because of its multi-layered structure. We also see the fundamental structure of ‘Understand, Know, and Do’ as epistemologically confused, and we doubt that this structure will assist teachers in clarifying their knowledge of epistemology to assist them as curriculum makers in providing the coherence and logic required for design for deep learning (McPhail, 2020a & 2020b). Also troubling is the almost total absence of acknowledgement of how learning actually happens, for example from the extensive cognitive science literature, and the relationship between this knowledge of cognitive development and the affordances of exposure to epistemically structured knowledge as fundamental to human cognitive development (Geary & Berch, 2016). This would involve, as Young (2020) suggests, that students develop ‘a relationship to knowledge’ by ‘becoming members, initially very junior members, of a range of subjects as “communities of enquiry” within which they engage with teachers and senior students as more knowledgeable members’ (p.16). We might expect to see aspects of this knowledge about knowledge more explicitly signposted in the recontextualizing principles and in the regulative and instructional discourses of a curriculum for the 21st century.

Notes

1. ‘Te Tiriti o Waitangi (in English, the Treaty of Waitangi), New Zealand’s founding document, was meant to be a partnership between Māori and the British Crown. Although it was intended to create unity, different understandings of the treaty, and breaches of it, have caused conflict. From the 1970s the general public gradually came to know more about the treaty, and efforts to honour the treaty and its principles expanded’ (Orange, 2023).
2. taonga—a treasure, or anything valued by Māori for example the Māori language.
3. te reo—the Māori language.
4. tikanga—customary practices.
5. mātauranga Māori—Māori knowledge.

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